

HW-001 (VER 1.0)

ENTERPRISE MOBILE PDA & BARCODE HARDWARE

🛒 BD 6,845.000 TOTAL PROCUREMENT

Enterprise Handheld PDA Mobile Computer, POS Barcode & QR Scanner Hardware Procurement Specification Agreement

Formal hardware specification, technical sizing, and Bill of Quantities (BOQ) for 22 units of Zebra TC21 Android Touch Computer Handheld PDAs and 7 units of Desktop Countertop 1D/2D QR Barcode Scanners for Popular Auto Spare & A/C Parts Co. W.L.L.

CLIENT ENTERPRISE

Popular Auto Spare & A/C
Parts Co.

HANDHELD PDA FLEET

22 x Zebra TC21 (TC210K)

COUNTER SCANNERS

7 x 1D/2D QR Scanners

TOTAL INVESTMENT

BD 6,845.000

Executive Hardware Procurement Mandate: To enable real-time mobility across central warehouse stock verification, 60/40 daily cycle counts, bin-location transfers, and floor sales cart handoff, Popular Auto Spare & A/C Parts Co. W.L.L. requires enterprise-grade rugged Android PDA computers and counter checkout scanners. Consumer smartphones fail in dusty automotive spare parts environments; therefore, the industry-standard **Zebra TC21 (TC210K) Touch Computer** and high-speed counter QR scanners are specified to guarantee 100% hardware-to-ERP barcode synchronization.

1.0 Zebra TC21 (TC210K) Touch Computer Handheld PDA Specifications (22 Units)

22 UNITS • BD 300.000 / UNIT



Zebra TC21 (TC210K) Android Enterprise Touch Computer

Rugged Industrial Handheld PDA with Integrated 1D/2D Imager & Wi-Fi/Bluetooth

Designed specifically for multi-shift automotive warehouse inventory scanning, mobile POS cart handoff, and inward goods verification (PVN). Features a high-speed SE4710 1D/2D barcode imager that reads damaged, curved, and plastic-wrapped barcodes in under 100 milliseconds.

ANDROID ENTERPRISE GMS

CORNING GORILLA GLASS

IP67 WATER/DUST PROOF

REMOVABLE 3100MAH BATTERY

SCAN ENGINE

1D/2D SE4710

Dedicated hardware imager with red laser aiming pattern for rapid barcode and QR reading in dark warehouse aisles.

CONNECTIVITY

Wi-Fi & Bluetooth

Dual-band 802.11ac Wi-Fi with fast seamless roaming across warehouse APs + Bluetooth 5.0 for mobile thermal belt printers.

BATTERY POWER

Rechargeable Li-Ion

High-capacity removable 3100 mAh battery with PowerPrecision technology supporting 10+ hours continuous multi-shift scanning.

DURABILITY

IP67 & 4ft Drop

Withstands repeated 4 ft./1.2 m drops to concrete and full immersion in water/dust; resistant to automotive grease and oils.

QUANTITY ALLOCATION

22 Nos (Total)

Allocated across Central Warehouse (Inward, Verification, Dispatch) and all Retail Branch floor salespersons.

INVESTMENT SUBTOTAL

BD 6,600.000

22 units @ BD 300.000 per unit, complete with battery, USB-C charging cable, and pre-staged Android ERP client.

For fixed counter billing, checkout speed, and cash drawer coordination across all Popular Auto Spare branch retail counters, 7 units of enterprise omnidirectional 1D/2D QR code scanners are provisioned:

SCANNER SPECIFICATION	CONFIGURATION / INTERFACE	FUNCTIONAL OPERATIONAL SCOPE & CHECKOUT INTEGRATION
 Omnidirectional 1D/2D & QR Imager	AUTO-SENSE & STAND	High-frequency CMOS imager capable of reading dense automotive part labels, QR codes on packing slips, and digital QR codes on customer phone screens at up to 60 scans/min.
 USB HID & Virtual COM Interface	PLUG-AND-PLAY DRIVERLESS	Zero-delay key-stroke emulation feeding barcode data directly into the React 18 / PHP POS billing window without requiring manual cursor clicks.
 Branch Counter Allocation	7 BRANCH POS DESKS	1 dedicated scanner deployed at each physical branch cash/billing desk, synced with receipt printers and RJ11 electronic cash drawers.

ERP MODULE	HARDWARE ASSIGNED	WORKFLOW EXECUTION & OPERATIONAL TASK
Module 1: PCode & Item Master	Zebra TC21 PDAs (22 Nos)	Initial barcode label generation verification and bin location tagging during master catalog setup.
Module 3: Store & Stock Control	Zebra TC21 PDAs (Warehouse Team)	Goods Inward Verification (PVN), 60/40 daily sampling stock counts with day-closing lock, and inter-branch dispatch scans.
Module 4: POS & Branch Sales	Zebra TC21 (Floor) + Counter Scanners (7)	Floor salespeople scan parts directly at shelves for 1-scan QR cart handoff; counter staff scan QR carts for instant invoice generation.
Module 8: Hardware & Offline Sync	All 29 Hardware Devices	Centralized device management, automated APK updates via Wi-Fi, and offline SQLite queuing during network dropouts.

To ensure smooth deployment without operational downtime, SaNDS Lab provides complete pre-delivery hardware staging, custom Android firmware provisioning, and warranty support:

 **Android Kiosk Lock & MDM Policy**

✓ **Dedicated ERP Kiosk Mode:** Locks Zebra TC21 devices strictly to the Popular Auto Spare Android ERP app, preventing personal browsing, social media, or gaming.

✓ **Optimized Wi-Fi Roaming:** Pre-configured 802.11r/k fast-roaming parameters for seamless handoffs across warehouse access points.

✓ **Barcode DataWedge Calibration:** Fine-tuned scan intent parameters for instant input into React input fields without latency.

 **1-Year Manufacturer Warranty & RMA**

✓ **Comprehensive 1-Year Hardware Warranty:** Covers manufacturing defects, motherboard issues, and scan-engine laser performance.

✓ **Rapid RMA Replacement Protocol:** Defective devices under warranty are diagnosed and routed through authorized regional service centers.

✓ **Battery Health Diagnostic:** Built-in PowerPrecision tools to monitor battery cycle counts and capacity degradation over time.

DEPLOYMENT STAGE	TIMELINE TARGET	ENGINEERING MILESTONE & DELIVERABLE DESCRIPTION
Stage 1: Procurement & Delivery	10 – 14 Working Days	Order placement with authorized Zebra distributor, customs clearance, and delivery to SaNDS Lab tech center.
Stage 2: Staging & Firmware Lock	3 – 5 Working Days	Serial number tracking, battery testing, OS update, Kiosk lock deployment, and ERP APK installation on all 22 PDAs.
Stage 3: Branch Deployment & Testing	3 – 4 Working Days	On-site distribution to Central Warehouse and retail branches, Wi-Fi pairing, cashier training, and live trial sign-off.

Enterprise Hardware Procurement Investment

Complete hardware package comprising 22 units of Zebra TC21 Android Touch Computer Handheld PDAs and 7 units of Desktop Counter 1D/2D QR Scanners, inclusive of pre-delivery staging, MDM configuration, and 1-year warranty.

BD
6,845.000
TOTAL HARDWARE
INVESTMENT

ITEM #	HARDWARE MODEL & DESCRIPTION	QTY (NOS)	UNIT COST (BHD)	TOTAL AMOUNT (BHD)
1.0	Zebra TC21 (TC210K) Android Touch Computer Handheld PDA Enterprise Android OS • 1D/2D SE4710 Imager • Wi-Fi • Bluetooth • Removable Li-Ion Battery • Gorilla Glass • IP67 Rugged	22 Nos	BD 300.000	BD 6,600.000
2.0	Desktop / Countertop 1D/2D & QR Code Barcode Scanner Omnidirectional High-Speed CMOS Imager • USB Interface • Auto-Sensing Handsfree Stand • Screen/Paper QR Scan	7 Nos	BD 35.000	BD 245.000
TOTAL ENTERPRISE HARDWARE PROCUREMENT VALUE:				BD 6,845.000

 **Payment & Procurement Terms:** Hardware procurement is executed under a standard 2-tranche schedule: **70% Advance Payment (BD 4,791.500)** upon formal agreement execution and PO issuance to distributor; **30% Balance Payment (BD 2,053.500)** upon physical delivery, pre-staging verification, and live branch handover.

5.0 Executive Multi-Party Digital Acceptance Console

CRYPTOGRAPHIC SIGN-OFF

By digitally signing below, all three authorized executive parties formally approve and execute the hardware equipment specifications, quantities (22 Zebra PDAs + 7 Counter Scanners), and the total procurement investment of **BD 6,845.000**.

CLIENT REPRESENTATIVE

Popular Auto Spare & A/C Parts Co.
Authorized Representative / Management

 Pending Signature

Awaiting Sign-off

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HARDWARE & SYSTEMS PROVIDER

SaNDS Lab Middle East W.L.L.
Head of Hardware Infrastructure / CTO

 Pending Signature

Awaiting Sign-off

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STRATEGIC PARTNER

UniGlobal Solutions
Senior Delivery Partner & QA Lead

 Pending Signature

Awaiting Sign-off

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